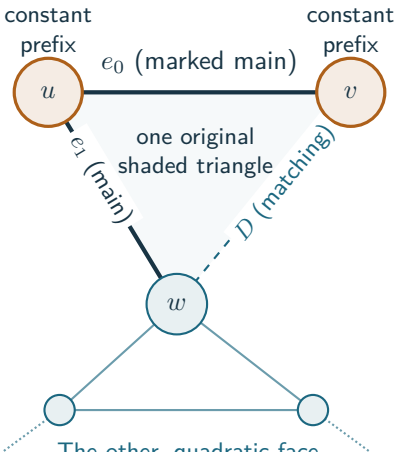
Two framing directions and a primitive neighboring cut

A patch of the white-face graph



Exactly two free face potentials

Subtract the common quadratic potential. Then

$$P_f = 0 \ (f \neq u, v), \qquad \dim \ker L = 2.$$

Two literal physical heads detect both

In the displayed orientation,

$$\begin{pmatrix} \delta h_{e_0} \\ \delta h_{e_1} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} P_u \\ P_v \end{pmatrix}, \qquad |\det| = 1.$$

Fix these two heads to obtain a primitive transverse slice. Recorded orientation reversals only change signs.

A source-coordinate statement

These two heads give source coordinates for the transverse directions. The positive face coefficient has a separate global saddle theorem.

Schematic dual face graph, not a knot diagram or a metric drawing.
Vertices represent white faces; edges correspond to original source wires.
Amber marks the two constant-prefix faces, blue the quadratic faces.